In [ ]:
```
import pandas as pd
dataset=pd.read_csv("startup_dataset.csv")
dataset
```

In [ ]:
```
dataset=dataset.drop(columns=["Unnamed: 0"])
dataset
```

In [ ]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

In [7]:
```
x=dataset[["RD Spend","Administration","Market Spend","State_Florida","State_New York"]]
y=dataset[["Profit"]]
print(x)
print(y)
```

	RD Spend	Administration	Market Spend	State_Florida	State_New York	Profit
0	165349.30	136897.90	471784.20	False	True	192261.93
1	162597.80	151377.69	443898.63	False	False	191792.16
2	153441.61	101145.65	407934.64	True	False	191050.49
3	144372.51	118671.95	383199.72	False	True	182902.09
4	142107.44	91391.87	366168.52	True	False	166188.04
5	131877.00	99814.81	362861.46	False	True	156991.22
6	134615.56	147198.97	127716.92	False	False	156122.61
7	130298.23	145530.16	323876.78	True	False	155752.70
8	120542.62	148719.05	311613.39	False	True	152211.87
9	123334.98	108679.27	304981.72	False	False	
10	101913.18	110594.21	229161.05	True	False	
11	100672.06	91790.71	249744.65	False	False	
12	93863.85	127320.48	249839.54	True	False	
13	91992.49	135495.17	252665.03	False	False	
14	119943.34	156547.52	256513.02	True	False	
15	114523.71	122616.94	261776.33	False	True	
16	78013.21	121597.65	264346.16	False	False	
17	94657.26	145077.68	282574.41	False	True	
18	91749.26	114175.89	294919.67	True	False	
19	86419.80	153514.21	0.10	False	True	
20	76253.96	113867.40	298664.57	False	False	
21	78389.57	153773.53	299737.39	False	True	
22	73994.66	122782.85	303319.36	True	False	
23	67532.63	105751.13	304768.83	True	False	
24	77044.11	99281.44	140574.91	False	True	
25	64664.81	139553.26	137962.72	False	False	
26	75328.97	144136.08	134050.17	True	False	
27	72107.70	127864.65	353183.91	False	True	
28	66051.62	182645.66	118148.30	True	False	
29	65605.58	153032.16	107138.48	False	True	
30	61994.58	115641.38	91131.34	True	False	
31	61136.48	152702.02	88218.33	False	True	
32	63408.96	129219.71	46085.35	False	False	
33	55494.05	103057.59	214634.91	True	False	
34	46426.17	157694.02	210797.77	False	False	
35	46014.12	85047.54	205517.74	False	True	
36	28663.86	127056.31	201126.92	True	False	
37	44070.05	51283.24	197029.52	False	False	
38	20229.69	65948.03	185265.20	False	True	
39	38558.61	82982.19	174999.40	False	False	
40	28754.43	118546.15	172795.77	False	False	
41	27893.02	84710.87	164470.81	True	False	
42	23641.03	96189.73	148001.21	False	False	
43	15505.83	127382.40	35534.27	False	True	
44	22177.84	154806.24	28334.82	False	False	
45	1000.33	124153.14	1904.03	False	True	
46	1315.56	115816.31	297114.56	True	False	
47	0.10	135427.02	0.10	False	False	
48	542.15	51743.25	0.10	False	True	
49	0.10	116983.90	45173.16	False	False	

```
9   149760.06
10  146122.05
11  144259.50
12  141585.62
13  134307.45
14  132602.75
15  129917.14
16  126993.03
17  125370.47
18  124267.00
19  122776.96
20  118474.13
21  111313.12
22  110352.35
23  108734.09
24  108552.14
25  107404.44
26  105733.64
27  105008.41
28  103282.48
29  101004.74
30   99937.69
31   97483.66
32   97427.94
33   96779.02
34   96712.90
35   96479.61
36   90708.29
37   89949.24
38   81229.16
39   81005.86
40   78240.01
41   77798.93
42   71498.59
43   69759.08
44   65200.43
45   64926.18
46   49490.85
47   42559.83
48   35673.51
49   14681.50
```

In [8]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
y_test
```

Out[8]:

	Profit
13	134307.45
39	81005.86
30	99937.69
45	64926.18
17	125370.47
48	35673.51
26	105733.64
25	107404.44
32	97427.94
19	122776.96

In [9]:
```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

In [10]:
```
from sklearn.ensemble import RandomForestRegressor
r=RandomForestRegressor(
     criterion='absolute_error', max_depth=None,
    max_features=1000, random_state=36, max_leaf_nodes=12,
    min_impurity_decrease=0.0
)
r.fit(x_train,y_train)
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
  return fit_method(estimator, *args, **kwargs)
```

Out[10]:

▾ RandomForestRegressor

▸Parameters

▸Fitted attributes

In [11]:
```
y_pred=r.predict(x_test)
y_pred
```

Out[11]:

```
array([135947.9996 ,  90320.2828 ,  98551.54685,  43705.64135,
       132715.42965,  38939.1742 , 114140.70525, 101620.57635,
        97708.65815, 130921.80825])
```

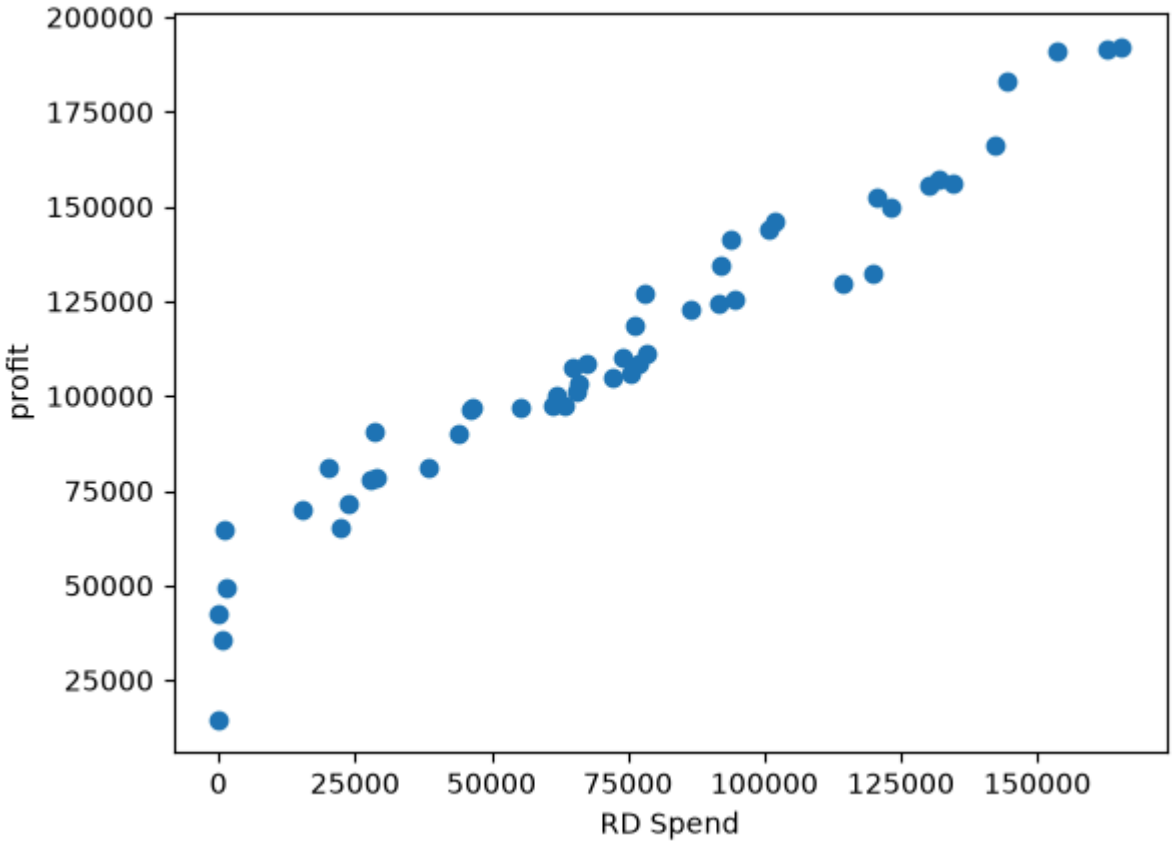
In [12]:
```
from sklearn.metrics import r2_score
```

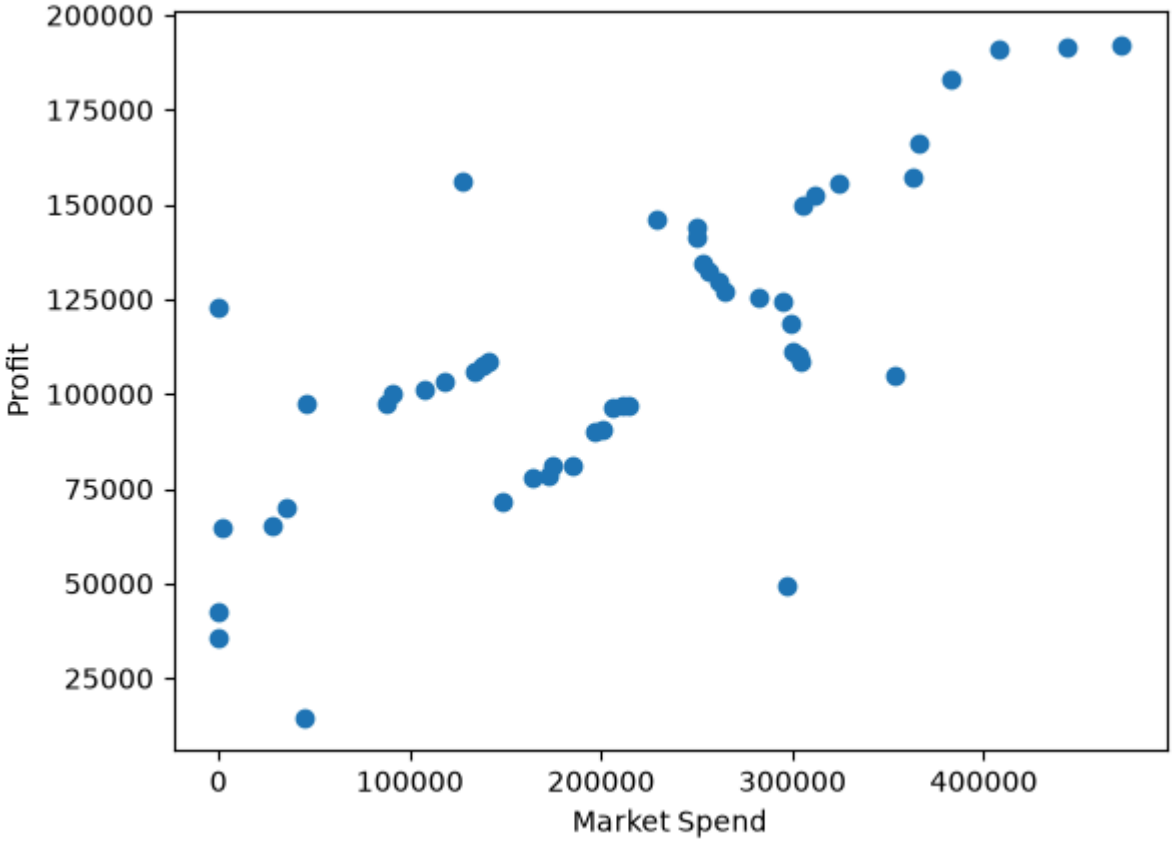
```
results=r2_score(y_pred,y_test)
results
```

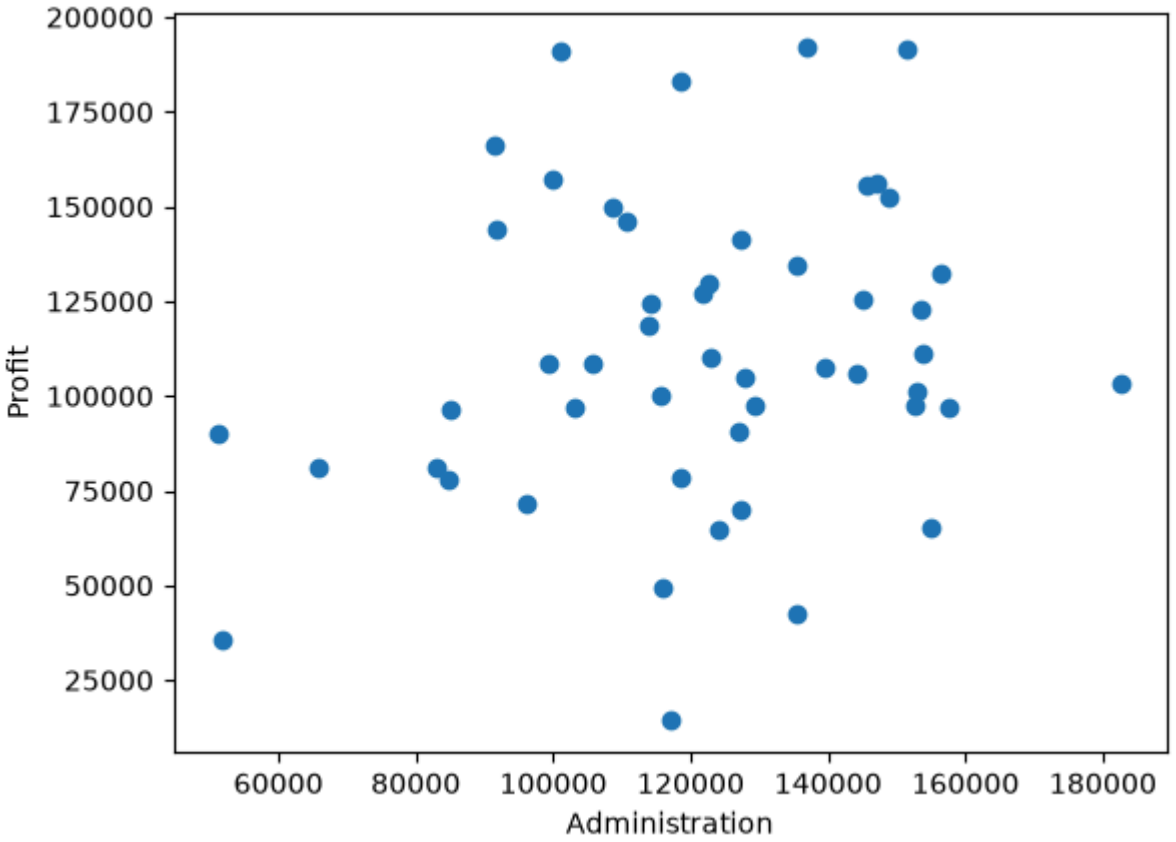
Out[12]:

0.9259867611891074

In [16]:
```
import matplotlib.pyplot as plt
plt.scatter(dataset["RD Spend"],y)
plt.xlabel("RD Spend")
plt.ylabel("profit")
plt.show()
plt.scatter(dataset["Market Spend"],y)
plt.xlabel("Market Spend")
plt.ylabel("Profit")
plt.show()
plt.scatter(dataset["Administration"],y)
plt.xlabel("Administration")
plt.ylabel("Profit")
plt.show()
```




200000
175000
150000
125000
100000
75000
50000
25000
Profit
0
100000
200000
300000
400000
Market Spend


200000
175000
150000
125000
100000
75000
50000
25000
Profit
60000
80000
100000
120000
140000
160000
180000
Administration

In [ ]: